ETHAN (YISONG) CAO

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EDUCATION

University of Virginia, School of Data Science
B.S. in **Data Science** | Concentration in **Machine Learning** & **Data Engineering** **Expected Graduation: May 2027**
M.S in **Data Science** | Concentration in **Artificial Intelligence (Admitted)** **Expected Graduation: August 2028**
Relevant coursework: Advanced Cloud Computing, Deep Learning, Mathematics for Machine Learning, Data Engineering

EXPERIENCES

Cisco Systems, Durham-Raleigh, NC (NASDAQ: CSCO)
Data Scientist Intern *May 2026 – August 2026*

- Post-Trained Qwen2.5-1.5B, Qwen3.5-4B, and Cisco's Antares-1B and Foundation-Sec-8B-Reasoning models with Jacobian-lens to develop a lightweight LLM Audit system against Prompt Injection attacks to Cisco's Agentic Systems.
- **GitHub: https://junewayne.github.io/Verbalizing-LLMs-For-AI-Defense/**
- Deployed an AI Search Engine with Sentence-Transformer embeddings and storing the vectors in Pgvector database, ranking search results with top cosine similarity scores between user query and escalation description embeddings.
- Architected and built a full-stack escalation tracking platform with FastAPI endpoints and Next.js frontend; Designed the PostgreSQL schema and deployed the application via OpenShift Containers with CI/CD pipelines set up with Jenkins.

Align Technology, San Jose, CA (NASDAQ: ALGN)
AI & Automation Intern *May 2025 – August 2025*

- Built and shipped a Multi-Agent RAG AI chatbot used by 20,000+ Align employees for market research analytics via Microsoft Azure AI Foundry, Copilot Studio, and Teams SDK in Python using LangChain.
- Enhanced retrieval accuracy on charts, dashboards, and table graphics by integrating Cohere-v4 multi-modal embedding model along with OCR (Optical Character Recognition) for text-heavy images; increased customer satisfaction by 116%+.
- Fine-tuned the o4-mini model on Azure AI Foundry via Supervised Fine-Tuning and Reinforcement Learning to improve chatbot response quality and retrieval relevance with document specific Question-Answers Pair.

RESEARCH PROJECTS

Spatial Artificial Intelligence Lab, Charlottesville, VA
Undergraduate Researcher *January 2026 – Present*

- Conducted mechanistic interpretability research on MeshXL, a 1.3B-parameter autoregressive 3D mesh transformer, discovering attention sink behavior and identifying an active/dormant attention head mechanism in mesh transformers.
- Engineered a PyTorch pipeline using forward hooks to extract attention matrices, weights, and hidden states; determine how active heads represent geometry by computing per-head feature deviations of attended mesh tokens against baseline metrics.
- Implemented a hidden-state retrieval analysis comparing each query token's active-head attention output to earlier mesh tokens' hidden states via cosine similarity across layers, revealing a consistent model bias toward adjacent geometry and primitives farther from the shape's centroid. In progress of investigating how to leverage findings for optimized generation.

Active Learning Lab, Charlottesville, VA
Undergraduate Researcher *December 2024 – May 2026*

- Built an unsupervised NLP pipeline analyzing 10,000+ LinkedIn early-career data science postings, scraping hourly job data and encoding descriptions into sentence embeddings to capture role content beyond surface-level job titles and descriptions.
- Reduced embeddings with UMAP and applied density-based HDBSCAN clustering to surface 45 distinct role groupings, then labeled each using class-based TF-IDF keyword extraction paired with LLM-generated categorical names.
- Tracked how roles evolve over time using Maximum Mean Discrepancy to detect month-to-month language drift and a bootstrap consistency metric to measure intra-cluster coherence, revealing that early-career data science is fragmenting into specialized "micro-professions."
- **Paper (Tentative): https://drive.google.com/file/d/1fwy6emwKNRC6io7qR5sn1kfe-yxhLq7Z/view?usp=sharing**
- **GitHub: https://github.com/JuneWayne/Latent-Representation-Job-Market/tree/main**

Visual Intelligence Lab, Charlottesville, VA
Research Intern *April 2025 – September 2025*

- Designed and trained a Physics-Aware Recurrent Convolutional (PARCv2) Neural Network via PyTorch to predict Advection & Diffusion behavior in 2D Shear Flow Fluid Dynamics Systems.
- Accelerated overall training time by 67%+ by implementing one third spatial down sampling and time step skipping, reducing per-epoch training time from 8+ hours to 6+ minutes.
- Optimized model rollout inference stability (reduce inference MSE by 81%+) by experimenting with Euler and RK4 integrators under the same model configurations and utilizing optimal checkpoint model weights via lowest validation loss.
- **GitHub: https://github.com/JuneWayne/PARCtorch/**

SKILLS

- **Programming Languages:** Python, SQL, Java, C++, HTML/CSS.
- **Skills:** PyTorch, Sci-kit Learn, LangChain, LangGraph, CrewAI, Git/GitHub, OpenAI API, Generative AI, Unix/Linux, Prefect, Relational & NoSQL Database, Hadoop, Spark, DuckDB, Kafka, Docker, TensorFlow, CUDA